1. Introduction

This week I learned about three types of machine learning: **Supervised Learning, Unsupervised Learning, and Reinforcement Learning**. I learned that each type learns in a different way depending on the type of data and the goal of the problem. We also practiced using machine learning with the Wine dataset, which helped me understand how a model can learn from data and make predictions.

2. Description of Experience or Topic

Background Information

Machine learning allows computers to learn patterns from data instead of being programmed with every possible rule. Supervised learning uses labeled data; unsupervised learning looks for patterns in data without labels, and reinforcement learning learns through rewards and penalties.

Specific Details

For this week's project, I worked with the Wine dataset. The dataset contains different measurements of wine, such as alcohol, malic acid, flavanoids, color intensity, and proline. I used these features to build a classification model that predicts the type of wine. I also experimented with different combinations of three features to see how changing the features affected the model's accuracy.

3. Personal Reflection

Thoughts and Feelings

At first, the different types of machine learning seemed a little confusing because they each have different purposes. After working with the dataset and running the models, I started to understand the differences better. Seeing the model make predictions from the data made the concepts easier for me to understand.

Analysis and Interpretation

One thing I learned is that the features selected for a model can make a difference in its performance. When I tried different features from the Wine dataset, I could compare the accuracy and see how the choice of data affected the results. This showed me that building a machine learning model is not only about choosing an algorithm. The data and features are also very important.

Connections to Theoretical Knowledge

This project connected directly to what I learned about supervised learning. Since the Wine dataset already has known wine classes, the model can learn from those labeled examples and use them to predict the class of new data. This helped me understand the difference between supervised learning and unsupervised learning.

4. Discussion of Improvements and Learning

Personal Growth

This project helped me become more comfortable working with machine learning data. I learned that it is important to experiment instead of assuming that one model or one group of features will always produce the best results.

Skills Developed

I developed skills in preparing a dataset, selecting features, splitting data into training and testing sets, training a Logistic Regression model, and evaluating its accuracy. I also became more comfortable using Python and libraries such as pandas and scikit-learn.

Future Application

I can apply what I learned to future projects involving classification and prediction. For example, machine learning could be used in healthcare to predict whether a patient may be at risk for a certain condition based on medical information. I can also use feature selection and model comparisons in future school projects to improve my results.

5. Conclusion

Overall, this week helped me better understand **Supervised, Unsupervised, and Reinforcement Learning** and how they learn from data in different ways. The project gave me practical experience with supervised learning and showed me how important feature selection is. I also learned that improving a machine learning model requires testing different approaches and analyzing the results. This experience gave me a better foundation for working with machine learning in future projects.